Solved Math Problems

solved problems

problems designed by yours truly

1. Show that Hermitian operators are normal operators; What is a normal operator; an operator A for which $AA^\dagger = A^\dagger A$; What is a hermitian operator: an operator B for which $B = B^\dagger$; Let's take a fixed, arbitrary operator B_f for which $B_f = B_f^\dagger$, and show that $B_f B_f^\dagger = B_f^\dagger B_f$;

$$B_f B_f^\dagger = B_f B_f = B_f^\dagger B_f \blacksquare$$

exercises

Exercise 2.1: Linear dependence

We are asked to consider the linear independence of the vectors

$$\begin{pmatrix} -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

In a 2-dimensional space ($\mathbb{R}^2$), any set of three or more vectors is linearly dependent. Therefore, this set of four vectors is linearly dependent.

As a proof by example, we can show one vector can be written as a linear combination of others. For instance:

$$\begin{pmatrix} -1 \\ 1 \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \beta \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

This gives us a system of two equations:

$$\begin{aligned} -1 &= \alpha + 2\beta \\ 1 &= 2\alpha + \beta \end{aligned}$$

Solving this system, we find $\alpha = 1$ and $\beta = -1$.

$$\begin{pmatrix} -1 \\ 1 \end{pmatrix} = (1) \begin{pmatrix} 1 \\ 2 \end{pmatrix} + (-1) \begin{pmatrix} 2 \\ 1 \end{pmatrix} \blacksquare$$

Exercise 2.2: Matrix Representation of an Operator

Let $A : V_0 \to V_0$ be a linear operator on the vector space spanned by the basis vectors $|0\angle$ and $|1\angle$. The action of the operator is given by:

$$A|0\angle = |1\angle \quad \text{and} \quad A|1\angle = |0\angle$$

The matrix representation of A in the basis $\{|0\angle, |1\angle\}$ is found by determining how A acts on each basis vector. Let the matrix be $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

- $A|0\angle = A\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \Rightarrow a = 0, c = 1$.
- $A|1\angle = A\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} b \\ d \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow b = 1, d = 0$.

Thus, the matrix representation of A is:

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

This matrix is known as the Pauli-X matrix, often denoted by σ_x. Show that all transformation whether they are rotating transformations or not, have at least one complex eigenvalue; Show this especially for rotating transformation;

Since virtually any input and output bases except precisely those pair of bases that are resultant from applying the exact same transformation to both the vector base input & output bases, could give rise to a different matrix representation of A, I leave the problem here to be trivially solvable. Say you can take $\{|0\angle, |1\angle\}$ as input basis & $\left\{\frac{|0\angle+|1\angle}{\sqrt{2}}, \frac{|0\angle-|1\angle}{\sqrt{2}}\right\}$ as output basis & you would get a different representation. Well, even if you take same input basis & change only one of the output basis vectors, you would still get a different matrix representation of A.

Exercise 2.2, rabbithole: Change of Basis

We have the same operator A (the Pauli-X matrix), but we want to find a new output basis such that its matrix representation is diagonal, specifically:

$$A' = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

This new basis, the eigenbasis of A, is composed of the eigenvectors of A. We can find them by solving the eigenvalue equation $A|v\angle = \lambda|v\angle$.

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \lambda\begin{pmatrix} a \\ b \end{pmatrix}$$

The characteristic equation is $\det(A - \lambda I) = 0$:

$$\det\begin{pmatrix} -\lambda & 1 \\ 1 & -\lambda \end{pmatrix} = \lambda^2 - 1 = 0 \Rightarrow \lambda = \pm 1$$

The eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = -1$.

- For $\lambda_1 = 1$:

  $$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow -a + b = 0 \Rightarrow b = a$$

  The normalized eigenvector is $|v_1\angle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}(|0\angle + |1\angle)$.
- For $\lambda_2 = -1$:

  $$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow a + b = 0 \Rightarrow b = -a$$

  The normalized eigenvector is $|v_2\angle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}}(|0\angle - |1\angle)$.

The new output basis is the set of eigenvectors $\{|v_1\angle, |v_2\angle\}$. The action of A on these basis vectors is:

$$A|v_1\angle = 1|v_1\angle \quad \text{and} \quad A|v_2\angle = -1|v_2\angle$$

In this new basis, the matrix representation of A is indeed

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

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